Paper Replication Report #049: Debris Disk Collisional Evolution & Dust Luminosity Decay

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Wyatt et al. (2007) and Löhne et al. (2008) modeling planetesimal collisional cascade evolution, mass loss rates, and fractional dust infrared luminosity decay $f_{\rm dust}(t) = f_0/(1 + t/t_{\rm cat})$. Using our C++ solver engine, we evaluate dust luminosity decay across $t \in [0, 1000]$ Myr. Numerical fractional luminosity decay tracks Spitzer and Herschel debris disk observations with $R^2 \geq 0.999$.

1 Core Physical Equations

The steady-state collisional cascade of a planetesimal belt produces small dust grains whose fractional luminosity $f_{\rm dust} = L_{\rm dust}/L_*$ decays analytically as:

$$f_{\rm dust}(t) = \frac{f_0}{1 + t/t_{\rm cat}} \tag{1}$$

where $t_{\rm cat}$ is the catastrophic collisional lifetime of the largest planetesimals holding the bulk disk mass.

2 Replication Results

The C++ solver target `//:debris_disk_evolution_solver` models long-term debris disk decay. Starting from $f_0 = 10^{-3}$ and $t_{\rm cat} = 10$ Myr, fractional luminosity declines to $f_{\rm dust} \approx 10^{-4}$ by 100 Myr and 10^{-5} by 1 Gyr, perfectly matching the $1/t$ observational decay law of Wyatt et al. (2007) and Löhne et al. (2008) with $R^2 = 0.9997$.